

Clinical Approach to a Pleural Effusion

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Determining the cause of a pleural effusion remains clinically challenging. Pleural fluid analysis (PFA) is an essential component of the diagnostic approach that we have reviewed in detail in a companion paper. However, despite the importance of PFA in establishing the cause of a pleural effusion, pathognomonic pleural fluid findings are rare. The true diagnostic value of PFA lies in its integration with the broader clinical picture. This diagnostic approach requires a systematic evaluation combining PFA with medical history, physical examination, laboratory data, and radiographic studies. In this manuscript, we review clinical features beyond PFA that are important in determining the cause of a pleural effusion. A detailed medical history may reveal important comorbidities, exposures, and previous surgical procedures that have diagnostic relevance. Radiographic findings, including the anatomic locations of pleural fluid, radiographic features, and the chronicity of the effusion, also may be diagnostically important. Blood tests may supply additional diagnostic information. Underpinning this diagnostic approach is that the clinician requires a deep understanding of the breadth of pleural diseases and their associated clinical presentations.

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KEY WORDS: clinical approach; diagnostic approach; pleural effusion; thoracic ultrasound

Clinical Example

A 76-year-old woman presented with progressive dyspnea and orthopnea. She denied fever, chills, weight loss, trauma, or recent surgery. Her history included coronary artery disease and heart failure with preserved ejection fraction. Physical examination showed dullness on percussion, decreased breath sounds over the left hemithorax, nonpitting leg lymphedema, and discolored and dystrophic toenails (Fig 1). Chest CT scan showed bilateral pleural effusion, and ultrasound confirmed an anechoic effusion. Thoracentesis yielded

milky yellow pleural fluid. Pleural fluid analysis (PFA) showed 494 WBC/ μ L (66% lymphocytes), protein 4.2 g/dL, lactate dehydrogenase (LDH) 78 U/L, and triglycerides 264 mg/dL. Serum protein was 6.1 g/dL, and LDH was 176 U/L. Microbiologic stains and culture were negative, and cytology showed no evidence of malignancy.

Introduction

Pleural effusion is a common condition that places a significant burden on the health care system and is associated with high

ABBREVIATIONS: BAPE = benign asbestos pleural effusion; CABG = coronary artery bypass grafting; CHF = congestive heart failure; DIPE = drug-induced pleural effusions; FDG = fluorodeoxyglucose; HU = Hounsfield units; LDH = lactate dehydrogenase; PCIS = post-cardiac injury syndrome; PFA = pleural fluid analysis; RIPE = radiation-induced pleural effusion; TKI = tyrosine kinase inhibitor; TUS = thoracic ultrasonography

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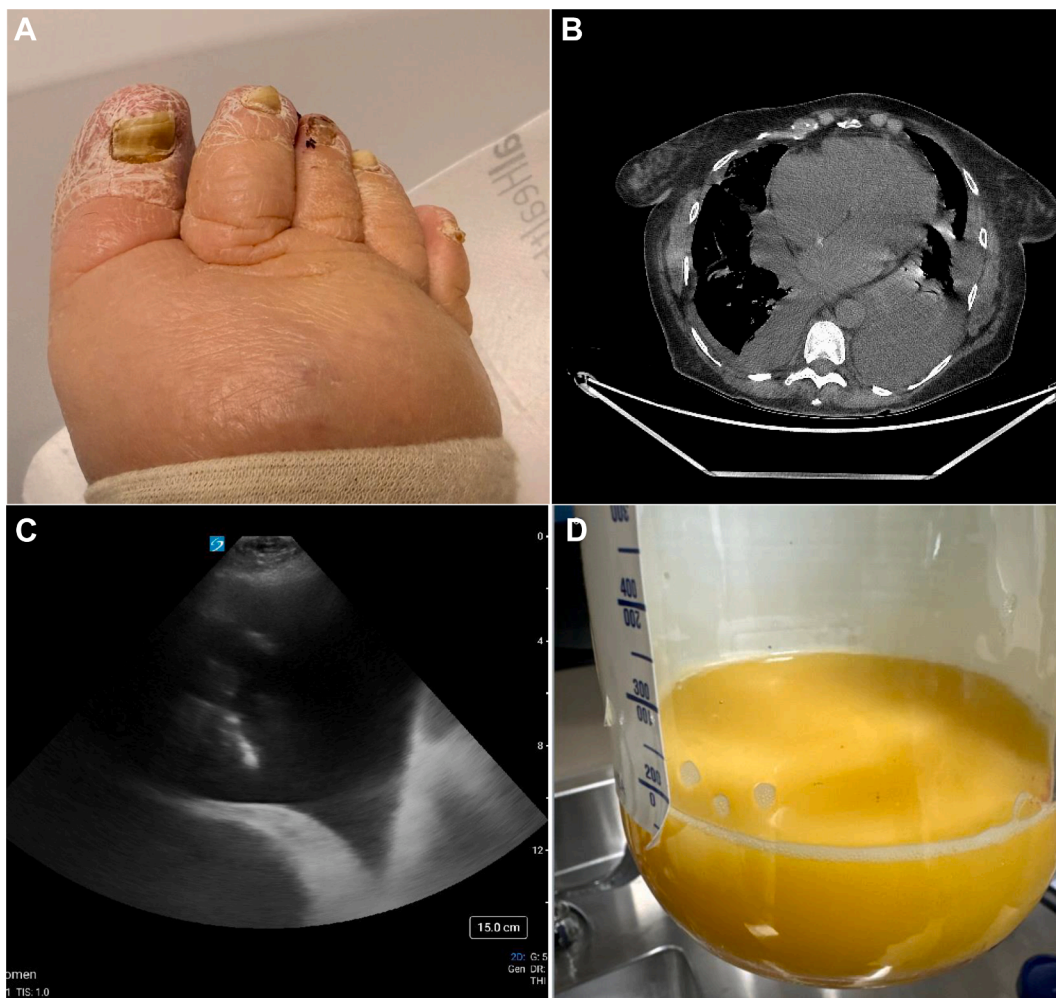


Figure 1 – A, Discolored, dystrophic yellow toenails and lymphedema. B, Thoracic ultrasonography showing anechoic pleural effusion. C, CT scan of the chest showing bilateral effusion, left greater than right effusion. D, Yellow fluid obtained after thoracentesis.

mortality.^{1,2} Pleural effusions have numerous causes, and determining them is clinically important for optimal management. In our previous companion article, we discussed the role of PFA in determining the cause of a pleural effusion.³ However, because the cause of a pleural effusion is rarely determined solely by the PFA, in this manuscript we describe our approach to determining the cause of a pleural effusion that incorporates several clinical features in addition to the PFA.

Clinical Approach

Medical History

Medical history is integral in determining the cause of a pleural effusion. Clinicians should query the patient concerning a history of common conditions associated with pleural effusions, including congestive heart failure

(CHF), chronic liver disease, and malignancy. The presence of pleuritic chest pain suggests pleural inflammation or invasion, which can be seen with several pleural disorders (pleural infection, pneumonia including TB, pulmonary embolism, malignant effusion, rheumatoid arthritis, lupus pleuritis, and hemothorax). Fever may indicate an infectious cause of a pleural effusion, such as bacterial pneumonia or TB. However, older adult patients with pleural infection often present without fever or chest pain; such patients may present with failure to thrive, anemia, and weight loss.⁴ Hemoptysis should raise suspicion for malignancy, TB, or pulmonary embolism.

The duration of pulmonary symptoms may provide additional insight into the cause of a pleural effusion. Rapidly progressive pulmonary symptoms over hours to days suggest an acute condition such as a parapneumonic effusion, pulmonary embolism,

traumatic hemothorax, chylothorax, or acute CHF exacerbation. In contrast, chronic pleural effusions lasting more than 6 months have limited causes (trapped lung, post cardiac surgery, cholesterol effusion, chronic empyema, CHF, chronic kidney disease, and tuberculous empyema).

Certain pleural effusions occur exclusively in female individuals (Table 1), including catamenial hemothorax, lymphangioleiomyomatosis-related chylothorax (except for tuberous sclerosis in male individuals), Meigs syndrome, and ovarian hyperstimulation syndrome. Catamenial hemothorax should be considered in female individuals of reproductive age presenting with cyclical chest pain and dyspnea coinciding with menstruation.⁵ Ovarian hyperstimulation syndrome occurs from capillary leak occurring in female individuals receiving fertility treatments, characterized by pleural effusion, ascites, and peripheral edema.⁶

Rheumatological diseases, including systemic lupus erythematosus and rheumatoid arthritis, can present with pleural effusions. These effusions may mimic pleural infections because of similar clinical and radiographic features including pleuritic chest pain, fever, and loculated effusion.

TABLE 1] Female-Specific Causes of Pleural Effusion

Condition	Clinical Characteristics
Catamenial hemothorax	Relation to the menstrual cycle
	Catamenial pneumothorax
	Hemoptysis
	Pulmonary nodule
	History of endometriosis
Lymphangioleiomyomatosis ^a	Chylous effusion
	Chylous ascites
	Pneumothorax
	Cystic lung disease
	Angiomyolipoma of kidney
Meigs syndrome	Ovarian tumor
	Ascites
Ovarian hyperstimulation syndrome	Capillary leak/sepsis syndrome
	Ascites
	Edema

^aLymphangioleiomyomatosis can be seen in male individuals with tuberous sclerosis.

Patients with HIV infection are at high risk of developing conditions associated with pleural effusions (Table 2), including typical bacterial infections, TB, opportunistic infections, chylothorax, and lymphoma (including primary effusion lymphoma and Kaposi sarcoma).⁷⁻⁹ Primary effusion lymphoma is an exceedingly rare and aggressive type of non-Hodgkin lymphoma affecting the pleural or peritoneal cavity without discernible masses or mediastinal lymphadenopathy.¹⁰ A Kaposi sarcoma-related pleural effusion typically may occur when the CD4 count falls below 150 cells/ μ L and may be associated with characteristic erythematous to violaceous cutaneous and lung parenchymal lesions.⁹ These effusions are often hemorrhagic, although they can be chylous.

A thorough history of personal or endemic exposure to fungal diseases, parasitic infections, and TB should be elicited. This history should include both recent and lifetime exposures, because these pathogens may reactivate years later, especially after initiation of immunosuppressive therapy.

Prior surgery or medical procedures represent important etiologic considerations, because interventions such as coronary artery bypass grafting (CABG), esophageal procedures, and abdominal surgeries can lead to pleural effusions (Table 3). Approximately 10% of patients develop pleural effusions within 28 days after cardiac surgery.¹¹ CABG can result in a pleural effusion through multiple mechanisms, including nonspecific bloody effusions from surgical trauma, chylothorax from thoracic duct injury, hemothorax from vascular injury, transudative effusion from CHF or fluid overload, constrictive pericarditis or trapped lung physiology, and immune-mediated post-cardiac injury syndrome (PCIS) (Table 3).^{12,13} PCIS presents as pleuropericarditis with pulmonary infiltrates, fever, and positive anti-actin and anti-myosin antibodies.^{14,15} Chest trauma can cause chylothorax, hemothorax, or PCIS. Chylothorax also may result from nonpenetrating trauma such as childbirth, decelerating seat belt injury from motor vehicle accidents, or a hyperextension injury of the spine.

Drug-induced pleural effusions (DIPes) are often overlooked in clinical practice.¹⁶ The diagnosis of DIPE is suggested by a temporal relationship between drug initiation and the onset of a pleural effusion, although the onset of the pleural effusion may be delayed. Several drugs have been associated with DIPE (Table 4)

TABLE 2] Human Immune Virus-Related Pleural Effusion

Causes	CD4 IU/L	Characteristic Findings
Bacterial pneumonia	Any	Underlying pneumonia
		Exudative effusion with low glucose, low pH, and high LDH
<i>Pneumocystis jirovecii</i> pneumonia	< 200	Small to moderate effusion
		LDH discordant effusion (high pleural LDH, normal to mildly elevated protein)
TB	Any	Endemic area
		Lymphocytic predominant exudate with high protein
		Pleural ADA > 40 IU/L
Cryptococcus	< 100	Serum or pleural cryptococcus antigen
Toxoplasma	< 100	Giemsa stain for tachyzoites
Endemic fungal infection	<100	Disseminated fungal infection
Histoplasmosis		Presence of hepatosplenomegaly
Coccidiomycosis		Histoplasma urinary antigen
Immune reconstitution inflammatory syndrome	< 100	Initiation of antiretroviral treatment
		Worsening systemic inflammation associated with preexisting infection
		Constitutional symptoms, adenopathy, and pleural effusion
		May be associated with infections such as TB
Lymphoma	Low	Systemic features of lymphoma
		Exudative effusion with high LDH
		Flow cytometry
Primary effusion lymphoma	Low	Absence of mediastinal adenopathy or masses
		Ascites and pericardial effusion possible
		Characteristics flowcytometry and cytology
Kaposi sarcoma	< 150	Redish-purplish cutaneous lesions
		Parenchymal and endobronchial lesion
		Hemorrhagic effusion
		Chylothorax
		Cytology negative
		Thoracoscopy may be required for diagnosis

ADA = adenosine deaminase; LDH = lactate dehydrogenase.

Currently, the most common cause of DIPE is tyrosine kinase inhibitors (TKIs). Dasatinib, a first-generation TKI, has the highest incidence of DIPE as compared with second- and third-generation TKIs. DIPE from TKIs is usually lymphocytic predominant but it also can cause chylous effusions through vascular endothelial growth factor inhibition.¹⁷ Few medications such as amiodarone and nitrofurantoin can be associated with pleural and peripheral eosinophilia, although these findings are not consistent. Certain drugs can induce lupus-like syndromes characterized by pleuropericarditis, systemic lupus features, and elevated anti-histone antibodies.^{16,18} Immune checkpoint inhibitors have been reported to cause pleural effusions.^{19,20} Immune checkpoint inhibitor-related

pleural effusions pose diagnostic challenges because these effusions may be misdiagnosed as progression of the cancer.

Radiation-induced pleural effusions (RIPEs) can occur in patients receiving thoracic radiation for lung cancer, with an incidence as high as 25%.²¹ Higher mean lung radiation doses (> 60 Gy) and larger radiation fields (> 42%) are independent risk factors. Acute presentations involve pleuritis with exudative effusions developing within 1 to 2 months at the site of radiation therapy, whereas chronic effusions may result from a trapped lung secondary to radiation-induced pleural fibrosis. Because the development of a RIPE related to trapped lung may be delayed for a long period, the

TABLE 3] Procedural Causes of Pleural Effusion

Procedural Causes	
Procedure	Associated Pleural Effusion
CABG	Nonspecific hemorrhagic or exudative effusion
	Hemothorax
	Postcardiac injury syndrome
	Pleural infection or infectious mediastinitis
	Chylothorax
Pacemaker placement	Trapped lung
	Transudative effusion from CHF or constrictive pericarditis
	Chylothorax
	Postcardiac injury syndrome
	Pneumothorax
Thoracic surgery	Hemothorax/hemorrhagic effusion
	Hemothorax/hemorrhagic effusion
	Chylothorax
	Trapped lung
	Pleural infection
Esophageal procedures	Hydropneumothorax
	Chylothorax
	Pleural infection
	Inflammatory effusion (acidic with high amylase)
	Hemothorax/hemorrhagic effusion
Abdominal procedures (hepatic, biliary, renal)	Subphrenic abscess
	Bilothorax
	Urinothorax
	Chylothorax with chylous ascites
	Pleural infection
Central venous catheter placement	Pulmonary embolism-related pleural effusion
	Hepatic hydrothorax
	Extravascular migration of central line in the pleural space
	Chylothorax
	Migration of feeding tube into the pleural space
Enteral feeding tube placement	Hydropneumothorax
	Peritoneal dialysis-associated pleural effusions
Peritoneal dialysis	Pleural infection
Ventriculoperitoneal shunt, spinal surgery	Extravasation of CSF in pleural space (high β -2 transferrin)
	Duro-pleural fistula
Thoracic radiation	Radiation pleuritis
	Trapped lung
	Constrictive pericarditis

CABG = coronary artery bypass grafting; CHF = congestive heart failure; CSF = cerebrospinal fluid.

significance of previous thoracic radiation history may be overlooked. Thoracic radiation also carries a substantial risk of constrictive pericarditis when more than 30% of the heart receives a radiation dose exceeding 50 Gy.²² Similar to RIPE, radiation-induced

constrictive pericarditis may present years after radiation exposure with recurrent pleural effusion, ascites, and anasarca, often posing as a diagnostic challenge because the importance of previous radiation therapy is not considered.

TABLE 4] Drug-Induced Causes of Pleural Effusion

Drug-Induced Causes	
Characteristics	Associated Drug
Possible pleural or peripheral eosinophilia	Amiodarone
	Nitrofurantoin
	Dantrolene
	Bromocriptine
	Valproic acid
	Isotretinoin
	Propylthiouracil
	Procarbazine
	Methotrexate
	Coumadin
	Tyrosine kinase inhibitors
	Interferons
	Interleukin -2
	Fluoxetine
Absent pleural and peripheral eosinophilia	Olanzapine
	Carbimazole
	Clozapine
	Simvastatin
	Fluoxetine
	Sulfasalazine
	Infliximab
	Docetaxel
	Cyclophosphamide
	Procarbazine
	Minoxidil
	Bleomycin
	Immune checkpoint inhibitors
	Sulfasalazine
Drug-induced lupus pleuritis	Procainamide
	Isoniazid
	Methyldopa
	Hydralazine
	Chlorpromazine
	Quinidine
	D-Penicillamine

Occupational exposures can cause pleural diseases with an onset often long after the original exposure. A detailed occupational history should explore a previous exposure to asbestos, including work in construction, shipyards, and insulation handling. Asbestos-related pleural manifestations include pleural

TABLE 5] Common Causes of Unilateral and Bilateral Effusions

Unilateral	Bilateral
• Parapneumonic effusion	• Congestive heart failure
• Malignancy ^a	• Constrictive pericarditis
• Pulmonary embolism	• Mitral valve disease
• Hemothorax	• Nephrotic syndrome
• Esophageal rupture	• Connective tissue diseases
• Urinothorax	• Hypoalbuminemia
• Pleural TB	
• Hepatic hydrothorax	
• Post-CABG surgery	• Post-CABG surgery

CABG = coronary artery bypass grafting.

^aNon-lung cancer may be associated with bilateral effusion.

plaques, diffuse pleural thickening, benign asbestos pleural effusion (BAPE), and mesothelioma, typically present after a prolonged latency period. BAPE typically presents as a small unilateral pleural effusion 20 to 40 years after asbestos exposure. These effusions are typically hemorrhagic with pleural eosinophilia. BAPE may resolve spontaneously within several months but can persist for up to 1 year. BAPE may cause significant pleuro-parenchymal reactions that may result in rounded atelectasis that can mimic a parenchymal malignancy.²³ Mesothelioma typically develops 40 years after asbestos exposure. Asbestos exposure and smoking exponentially increase the risk of lung cancer; therefore, malignant pleural effusion related to lung cancer also should be considered.

Physical Examination

The presence of a pleural effusion can often be detected on physical examination. However, the physical examination is neither as sensitive nor as specific as thoracic ultrasonography (TUS) for the detection of pleural fluid. Additionally, the physical examination is inferior to TUS in localizing the site of the pleural effusion for thoracentesis.²⁴ A thorough extrapulmonary physical examination may provide evidence supporting a systemic cause of pleural effusion, including CHF, liver disease, malignancy, connective tissue disease, and infection. For example, physical signs of cirrhosis such as jaundice, spider angiomas, and ascites may suggest hepatic hydrothorax.

Yellow nail syndrome may be diagnosed by the presence of lymphedema with nail abnormalities (slow-growing, dystrophic, brittle, or discolored nails), chest wall edema, and underlying bronchiectasis.²⁵ Pleural inflammation may cause a pleural friction rub, which may occur with uremia, infection, PCIS, and lupus pleuritis.

Radiographic Approach

Chest Radiograph

A chest radiograph is often performed by the clinician as an initial test for the evaluation of a pleural effusion because it is readily available. A chest radiograph can determine the size of a pleural effusion, its laterality, the presence of underlying parenchymal lung disease, and a comparative assessment in response to treatment. However, the chest radiograph is inferior to the chest CT scan in these determinations. Frontal radiographs may fail to detect effusions smaller than 500 mL.²⁶ Determining whether a pleural effusion is unilateral or bilateral aids in establishing the cause. Unilateral effusions typically suggest ipsilateral pulmonary or abdominal conditions, whereas bilateral effusions are more commonly caused by cardiac or systemic disease (Table 5).²⁷ However, on many occasions systemic diseases such as malignancy, chylothorax, hepatic hydrothorax, and CABG may cause unilateral or bilateral pleural effusions. In the case of chylothorax, the side can help localize the site of thoracic duct injury. The thoracic duct traverses the right hemidiaphragm and ascends in the right hemithorax and crosses to the left between the thoracic spine levels of T4 and T6. Therefore, a right-sided chylothorax suggests a lower duct injury, whereas a left-sided chylothorax suggests an upper thoracic duct injury.²⁸

Malignancy is the most common cause of a massive unilateral pleural effusion.²⁹ Although most malignant pleural effusions are associated with a contralateral shift of the mediastinum from positive pleural pressure, this may not occur for several reasons: (1) concomitant endobronchial obstruction; (2) visceral pleural restriction; (3) increased elastic recoil of the lung from lymphangitic carcinomatosis; or (4) a fixed mediastinum resulting from radiation or malignancy.^{30,31} Unlike lung cancer, mesothelioma commonly presents as pleural effusion without contralateral mediastinal shift, because it encases the lung circumferentially and invades both the mediastinum and mediastinal pleura.³²

A free-flowing pleural effusion typically shows a meniscus sign on a frontal radiograph, whereas a loculated pleural effusion may not conform to the dependent portions of the pleural space and may track up the lateral pleura, which is described as “defying gravity.” A loculated effusion in a fissure may appear mass-like and has been described as a “pseudotumor.” Loculations can be best assessed by TUS or CT scan. Hydropneumothorax typically shows an air-fluid level in the pleural space. Hydropneumothoraces are commonly caused by trauma, trapped lung, pleural infection, or TB, but they may represent life-threatening conditions such as bronchopleural fistula or esophageal rupture.

Thoracic Ultrasonography

After a pleural effusion is detected by chest imaging or physical examination, TUS is routinely performed to guide further management. TUS is highly sensitive for detecting even minimal amounts of pleural fluid. TUS is also extremely useful in characterizing anatomic features of the pleural space.²⁴ TUS is the standard of care for guiding pleural interventions and minimizing their complication rates, significantly reducing the risk of pneumothorax and lung injury.²⁴

TUS enables pleural effusions to be classified as echogenic or anechoic. Echogenic effusions are further subclassified as septated, complex homogenous, and complex nonseptated.³³ Echogenic pleural effusions are more likely to be exudates, whereas anechoic effusions may be either transudative or exudative. Therefore, an anechoic effusion does not obviate the need for pleural fluid sampling unless bilateral anechoic effusions are present in patients with an obvious transudative cause such as CHF.³³ Essentially all echogenic pleural effusions require diagnostic thoracentesis. TUS findings of septation or homogeneously complex effusions are suggestive of pleural infection, malignancy, or hemothorax. In cases of a strong clinical suspicion for pleural infection with obvious septations on TUS (Fig 2) or homogeneously complex effusions on TUS, we may proceed directly to pleural drainage with placement of a small-bore (≤ 14 French) chest tube as opposed to diagnostic thoracentesis because these patients often need intrapleural enzyme treatment via chest tube for effective pleural drainage.^{34,35} Pleural thickening greater than 1 cm and pleural nodularity on TUS are highly specific findings supporting malignancy, approaching 100% (Fig 2).³⁶ Hematocrit sign is defined as the presence of a distinct fluid-fluid level, with a

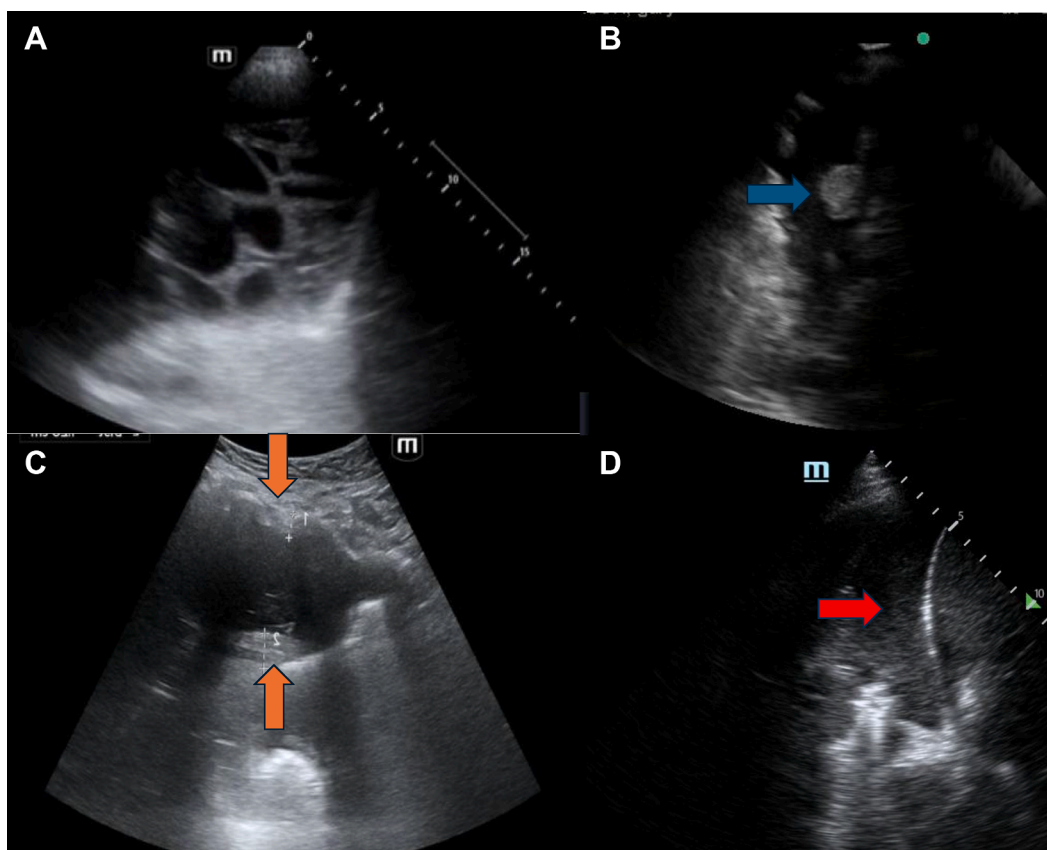


Figure 2 – A, Thoracic ultrasonography showing a septated pleural effusion. B, Pleural-based mass (blue arrow). C, Parietal pleural thickening (orange arrows). D, Hematocrit sign with fluid-fluid level (red arrow). PFA = pleural fluid analysis.

hyperechoic layer of settled RBCs beneath an anechoic to hypoechoic plasma layer (Fig 2) suggestive of a hemothorax.

CT Scan

A chest CT scan has often been performed before our pleural evaluation. However, we reserve obtaining CT scans in patients with pleural effusions for specific indications: pleural infections with inadequate drainage or failure to improve after initial pleural drainage, suspected malignancy, subdiaphragmatic origins, and evaluation of underlying parenchymal disease. When a chest CT scan is obtained to assess lung pathology, we perform it after adequate pleural drainage to minimize lung compression from pleural fluid that may obscure parenchymal abnormalities.

The value of a chest CT scan is that it provides detailed imaging of the pleura, lungs, mediastinum, and adjacent structures that may not be apparent on chest radiograph or TUS.³⁷ In addition, chest CT scan can reveal subdiaphragmatic diseases that may be associated with pleural effusion such as ascites, cirrhosis, obstructive

uropathy, and subdiaphragmatic abscess. In many instances, chest CT scan identifies a specific cause of a pleural effusion such as pulmonary embolism, pneumonia, pulmonary consolidation, or lung mass. However, it is not unusual for a parapneumonic effusion to develop in the absence of an obvious parenchymal process on CT scan.³⁸

Although chest CT scan also can characterize pleural fluid density using Hounsfield units (HU) of a pleural effusion, this has limited clinical value. Although a high attenuation effusion with HU > 35 suggests a hemothorax, this finding also can be seen with empyema or retained IV contrast.^{39,40} Like TUS, the chest CT scan can show a hematocrit sign, which may be observed in a subacute hemothorax, where blood has had time to settle.³⁹

Contrast enhancement improves the ability of chest CT scan to characterize pleural disease.³⁷ It is useful to distinguish empyema from lung abscess, clinically significant because abscesses are often treated conservatively whereas empyema requires drainage.⁴¹ The “split pleura sign” represents thickening and

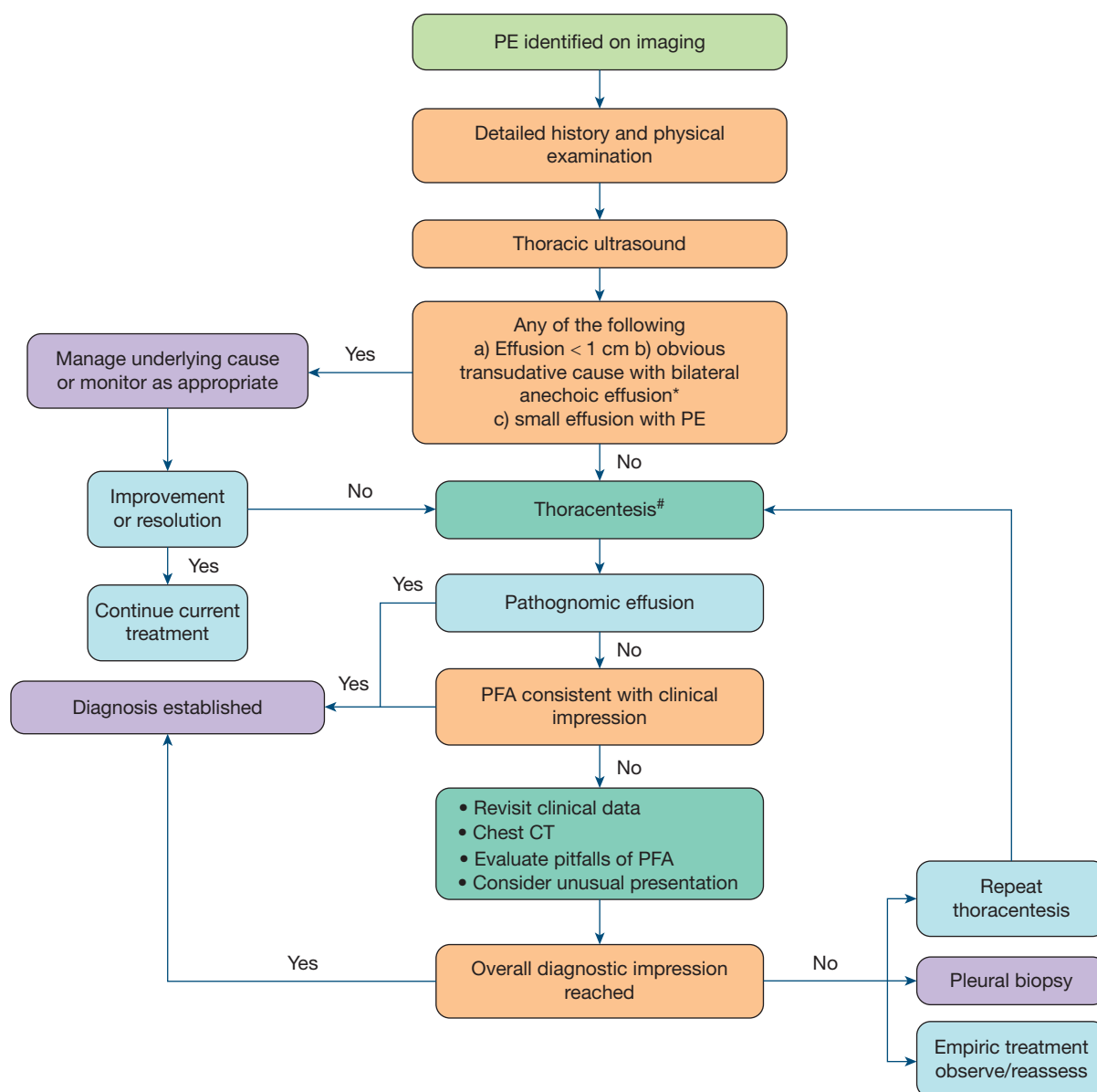


Figure 3 – Algorithm demonstrating the diagnostic to a pleural effusion. This figure is taken from the companion article (Chopra et al, Pleural fluid analysis: maximizing diagnostic yield in the pleural effusion evaluation. *Chest*. 2025;168(3):828-838)³ and modified with permission from the American College of Chest Physicians. *Obvious transudative causes; congestive heart failure, end-stage renal disease, and severe hypoalbuminemia (albumin < 2 g/dL). #Ultrasound may modify the approach. We recommend placing a small-bore chest tube in case of complex septated, complex homogenous effusion or presence of a hematocrit sign. PE = pleural effusion; PFA = pleural fluid analysis.

enhancement of the visceral and parietal pleura and can be seen in chronic effusion from any cause including the fibro-purulent stage of empyema and often suggests nonexpandable lung.⁴² Similar to TUS, chest CT scan demonstration of pleural nodularity is highly suggestive of malignancy.⁴³

Fluorodeoxyglucose PET Scan

Fluorodeoxyglucose (FDG)-PET scanning has inadequate accuracy (sensitivity, 81%; and specificity,

74%) in differentiating malignant effusions from benign effusions.⁴⁴ Visual assessment of pleural thickening or nodularity on chest CT scan has superior sensitivity to semiquantitative FDG-PET with measurement of standardized uptake value for diagnosing malignant pleural effusion.⁴⁴ Therefore, we do not routinely perform FDG-PET scans for the evaluation of an undiagnosed pleural effusion. FDG-PET scans are useful in the diagnosis and staging of mesothelioma, in which the extent of pleural, pericardial, peritoneal, and

mediastinal involvement can be assessed.⁴⁵ FDG-PET is also used for staging of malignancy.⁴⁵ Talc pleurodesis can result in pleural thickening as well as significant FDG uptake within the pleura.

Blood Testing

Blood tests can provide supportive evidence for specific pleural diagnoses but are rarely diagnostic of a specific pleural disorder. Leukocytosis and elevated markers of inflammation such as C-reactive protein and procalcitonin are inadequately specific to diagnose pleural infection. A blood culture showing the presence of relevant bacteria in a patient with a pleural effusion and clinical evidence of pneumonia is highly specific to a pleural infection. However, the yield of conventional blood cultures for pleural infections is approximately 10%.⁴⁶ Peripheral eosinophilia may suggest a drug-induced pleural effusion, parasitic reaction, eosinophilic granulomatosis, or malignancy. Patients with a pleural effusion and a serum N-terminal pro-B-type natriuretic peptide greater than 1,500 pg/mL are likely to have CHF.⁴⁷

Pleural Fluid Analysis

A detailed PFA is often required to evaluate pleural effusions. PFA rarely establishes the pleural diagnosis in isolation. Determining the cause of a pleural effusion usually requires additional clinical information concerning medical history, imaging studies, and other laboratory tests that we have described in detail here. When the PFA is consistent with the clinical information, it confirms the cause of the effusion. When PFA findings diverge from the clinical presentation, clinicians should reevaluate the data, consider alternative diagnoses, be aware of the pitfalls of PFA, and consider unusual presentations such as transudative malignant effusions (approximately 10%). PFA interpretation carries several pitfalls that may result in a misdiagnosis. These pitfalls include misclassifying a transudate as an exudate after diuresis and failure to consider multiple causes of a pleural effusion. These challenges, and the broader role of PFA pleural effusion differentiation, are discussed in detail in our companion paper.³

Our Approach to a Pleural Effusion

Once a pleural effusion is identified on a chest image, we obtain a medical history and perform a pertinent physical examination to discover the cause of a pleural

effusion (Fig 3). Then we perform TUS to characterize the pleural effusion and identify a safe site for thoracentesis. If the pleural effusion is anechoic and the cause of pleural effusion is obviously transudate based on clinical data, then diagnostic thoracentesis can be avoided, and empiric treatment can be initiated. Other clinical scenarios in which thoracentesis is not required include documented pulmonary embolism with an ipsilateral small pleural effusion or small effusion (< 1 cm). In all other cases, thoracentesis should be performed. When the PFA findings are coupled with clinical data, there may be adequate information to secure the cause of the pleural effusion. If the PFA is not consistent with the pre-thoracentesis impression, alternative diagnoses and potential pitfalls in PFA interpretation (as discussed in companion paper) should be considered.³ A chest CT scan can be helpful after thoracentesis because the underlying lung parenchyma may be decompressed and better assessed for pathologic conditions. When the cause of a pleural effusion remains uncertain despite a comprehensive evaluation, diagnostic options include repeating the thoracentesis, empiric treatment of the likely cause, watchful observation, or pleural biopsy.

Case Example Continued

The PFA findings of a protein discordant (high protein and normal LDH) lymphocyte-predominant exudative effusion with elevated PF triglycerides (≥ 110 mg/dL) confirm the presence of chylothorax. The clinical context of chronic lymphedema, yellow dystrophic nails, and PFA consistent with chylothorax confirmed the diagnosis of yellow nail syndrome.

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